

ETABLIX

INTEGRATEDSITESERVICES·PARTOFGROUPENSEYA

EMPLOYER'S REQUIREMENTS

THE TWELVE APPENDICES

P2 — Modular Accommodation

Client	Marrowbridge Infrastructure Ltd
Project	Project NORTHREACH — Workforce Accommodation Village
Parent document	NR-ER-P2 Rev B — Employer's Requirements, P2 Modular Accommodation
Document reference	NR-ER-P2-APX
Revision	A
Date	September 2026
Status	Issued for tender

Eight of the twelve are the Employer's own documents and are written here in full. Four are produced by others — the fire engineer, the planning authority, the environmental consultant, the highway authority. Those are issued as controlled insertion sheets stating what must be inserted, by whom, what this document depends on it for, and WHAT A TENDERER PRICES UNTIL IT ARRIVES. An appendix that says only "to be issued" transfers an unpriced risk to a tenderer, who will either price its worst case or qualify its return — and both defeat the evaluation.

Contents

Register of appendices

Every appendix, who authors it, its status, and the clauses of the Employer's Requirements that depend on it. A clause that depends on an appendix which is not yet issued is listed here rather than left for a tenderer to discover.

	Appendix	Authored by	Status	Depended on by
1	Activity schedule, in the structure to be priced	Employer	Issued in full	Section 12
2	Fire strategy	Employer's fire engineer	INSERTION SHEET — the strategy is a separate document	4.5, 4.15, 4.26, IF-11
3	Planning consent, conditions, and the Employer's site rules	Planning authority / Employer	Decision notice INSERTED; site rules issued in full	7.4, 14.2, 14.5, IF-12
4	Construction Environmental Management Plan	Employer's environmental consultant	INSERTION SHEET	10.6
5	Approved external colour and finish range	Employer	Issued in full	4.3, 4.22, room data sheets
6	Bed demand curve, occupancy programme and sectional completion dates	Employer	Issued in full	7.2, 18.1, 18.2, 4.12
7	Abnormal load route assessment	Highway authority / Employer's consultant	INSERTION SHEET — DRAFT, INCOMPLETE, and the risk is the Employer's	7.3, 11.1, 18.3
8	Design loadings, site exposure, and service supply characteristics	Employer	Issued in full	4.2, 4.4, 4.24, 4.25, 4.27
9	Overheating criterion and the Employer's comfort brief	Employer	Issued in full	Section 6, room data sheets
10	Form of collateral warranty	Employer's legal adviser	CONTENT SCHEDULE issued; the form itself is drafted by lawyers	13.6
11	Asset information requirements, data schema and the CDE	Employer	Issued in full	15.3, 15.4, IF-14, 4.24
12	Social value framework and the travel-to-work area	Employer	Issued in full	16.5

The four insertion sheets are not placeholders. Each one states the basis a tenderer prices on until the document arrives, and what happens if the issued document differs from that basis. Read them before pricing.

Appendix 1 Activity schedule

This is the structure the tender is priced in. A tenderer who alters it produces a return that cannot be compared with the others, and it will be marked accordingly.

Rates are inclusive of everything necessary to complete the item to the Employer's Requirements — labour, materials, plant, temporary works, supervision, testing, commissioning, overheads and profit — except where an item is expressly stated to be priced separately. A rate left blank is a rate of nil and the item is deemed included elsewhere; a tenderer who intends it to be excluded shall list it as a departure under 13.4 rather than leaving it empty.

Rules of pricing

Quantities	Quantities are the Employer's and are firm. Remeasurement applies only where this document expressly permits it. A tenderer that disagrees with a quantity shall state the quantity it has priced, in the Schedule of Departures, rather than adjusting the rate to compensate.
Rates for change	The all-in rates required at 17.2 are priced at item M and are used to value every variation for the life of the contract. They are competed here because this is the only moment at which competition exists.
Provisional sums	There are none. Where the Employer cannot define a scope it is excluded and named, per 1.4. A provisional sum is a decision deferred to a point at which it can no longer be competed.
Currency and fluctuation	Sterling. Fixed for the contract period. An indexation mechanism is a departure under item N with the index and base date stated.
The sample room	Item K is priced and programmed in advance of the main delivery. It is not a proportion of item B and shall not be priced as one.

The schedule

Item	Description	Unit	Qty	Note
A	DESIGN, CALCULATIONS AND APPROVALS			
A1	Design development to the responsibility matrix at 13.3, including the Contractor's Proposals	Sum	1	Includes every submission listed at 13.5
A2	Structural design, connections, lifting points and the temporary condition at 4.27	Sum	1	
A3	Thermal calculations, thermal bridging and the TM59 overheating model	Sum	1	Appendix 9
A4	Acoustic design and the flanking demonstration at 4.6	Sum	1	
A5	Drainage design and the trap seal demonstration at 4.23	Sum	1	Congested-use loading
A6	Maximum demand calculation, discrimination study and earthing design	Sum	1	4.24
A7	Water design, legionella written scheme of control and the flushing regime	Sum	1	4.11, 4.12
A8	Fire alarm design, cause and effect matrix and the BS 7273-4 door schedule	Sum	1	4.15, 4.18, 4.26
A9	Lightning protection risk assessment to BS EN 62305-2	Sum	1	4.17 — priced whether or not protection results
A10	Relocation method statement and the schedule of consumed components	Sum	1	13.7
A11	Collateral warranties — Contractor and each designing sub-contractor	Nr	6	Appendix 10
B	MANUFACTURE — BEDROOM MODULES			
B1	RT-01 standard bedroom with ensuite shower room	Nr	199	Rate per room, complete to the room data sheet

B2	RT-02 accessible bedroom with ensuite wet room	Nr	12	Rate per room
B3	RT-03 senior / long-stay bedroom	Nr	14	Rate per room
B4	Circulation, stairs and the module envelope serving the above	Sum	1	Not to be spread into B1–B3
C	MANUFACTURE — AMENITY, RECEPTION AND SHELL			
C1	Reception, management office and resident amenity space	m ²	420	
C2	Kitchen and dining building shell, to receive P3	m ²	610	IF-04. Shell only — fit-out is P3
C3	Laundry, drying and stores	m ²	180	
C4	Plant enclosures, acoustically treated, with the access at 4.14	Sum	1	
D	TRANSPORT AND DELIVERY			
D1	Delivery to site, bedroom modules	Nr	88	Modules, not rooms. State the module count assumed
D2	Delivery to site, amenity and shell modules	Nr	26	
D3	Abnormal load escorts, permits and route compliance	Sum	1	Appendix 7 — route assessment incomplete, risk with the Employer per 18.3
E	CRANEAGE, SETTING AND CONNECTION			
E1	Craneage, lift planning to BS 7121-1 and the Appointed Person	Sum	1	4.21
E2	Setting, levelling and inter-module connection	Nr	114	Per module set
E3	Weathering, joint sealing and the joint mock-up test	Sum	1	4.22
E4	Connection to the P1 termination points	Nr	8	Per block, all four services
E5	Temporary works design, independent check and permits	Sum	1	4.21 — Category 2 minimum
F	INTERNAL FIT-OUT AND FIXED JOINERY			
F1	Fit-out, RT-01	Nr	199	Where not included in B1 — state which
F2	Fit-out, RT-02	Nr	12	
F3	Fit-out, RT-03	Nr	14	
F4	Circulation and amenity finishes, wall and corner protection	Sum	1	4.7, 4.9
F5	Signage and wayfinding to the scheme at 4.20	Sum	1	Including the viewing-distance calculation
G	MECHANICAL AND ELECTRICAL			
G1	Heating and controls to the duty in the room data sheets	Sum	1	State the resulting connected load per block
G2	Ventilation and ensuite extract	Sum	1	
G3	Hot and cold water, TMVs, boosting and metering	Sum	1	4.11, 4.25
G4	Above-ground drainage within the modules	Sum	1	4.23
G5	Electrical distribution, protection and metering	Sum	1	4.24
G6	Lighting and emergency lighting	Sum	1	4.16
G7	Fire detection and alarm, Category L1	Sum	1	4.15, 4.26
G8	Access control, offline battery locks and the key hierarchy	Sum	1	4.18 — fail-secure to bedrooms, fail-safe to escape doors
G9	Structured cabling and the wireless network	Sum	1	Coverage per section 6
G10	CCTV containment, cabling and power to the P5 termination points	Sum	1	4.19 — cameras are P5

G11	Lightning protection installation, where the assessment requires it	Sum	1	Price nil if the assessment concludes none is required, and say so
H	COMMISSIONING, TESTING AND DEMONSTRATION			
H1	Commissioning, every outlet and every extract measured and recorded	Nr	225	Per room. 8.5 — summaries are rejected
H2	Pre-completion acoustic testing	Nr	23	One room pair in ten
H3	Air permeability testing, ATTMA TSL2	Nr	3	One completed block in three, joints included
H4	Weathertightness testing — joint mock-up and site hose test	Sum	1	4.22
H5	Demonstration of cause and effect, access control and the wireless heat map	Sum	1	8.6
I	HANDOVER DELIVERABLES AND TRAINING			
I1	As-installed drawings, model and O&M manuals	Sum	1	15.2
I2	Asset register in the schema at Appendix 11	Sum	1	IF-14
I3	Registers — doorsets, firestopping, keys, valves	Sum	1	
I4	Training for P5, two recorded sessions	Sum	1	
J	SPARES AND CONSUMABLES — PRICED SEPARATELY			
J1	Twelve months' consumables from the last sectional completion	Sum	1	19.5 — stated separately so the Employer can decline it knowingly
J2	Critical spares holding, criticality 1 assets	Sum	1	19.5
K	SAMPLE ROOM			
K1	One complete RT-01, fully finished and serviced, in advance	Nr	1	8.3, 8.7 — not a proportion of B1
L	OPTIONS — PRICED, NOT INCLUDED IN THE LUMP SUM			
L1	Window-open heating interlock, as a priced option	Sum	1	4.10
L2	Self-testing emergency lighting	Sum	1	4.16
L3	Improvement on the air permeability figure, per 1.0 m ³ /(h·m ²) below 7.0	Rate	1	Evaluated — section 6
M	RATES FOR THE VALUATION OF CHANGE			
M1	All-in hourly rate, by trade and grade, site	Rate	—	17.2 — list every trade and grade to be employed
M2	All-in hourly rate, by trade and grade, factory	Rate	—	17.2
M3	Percentage addition for overheads	%	—	
M4	Percentage addition for profit	%	—	
M5	Percentage addition on sub-contracted work	%	—	
M6	Craneage, per shift and per hour	Rate	—	
N	DEPARTURES FROM THESE EMPLOYER'S REQUIREMENTS			
N1	Each departure listed and priced individually, plus or minus	Nr	—	13.4 — a departure not listed is not accepted

Blank unit and quantity mark a section heading. Every other line is priced.

Appendix 2 Fire strategy

Authored by	The Employer's fire engineer
Status	To be inserted. Revision C is expected before the clarification deadline.

2.1 What the Employer's Requirements depend on it for

- The evacuation strategy and zoning decisions at 4.26 — the Employer has stated them and the fire engineer's strategy must confirm or vary them.
- Compartmentation and the REI 60 requirement at 4.6 and section 6.
- Escape route widths and travel distances, which set the corridor width at 4.7 and therefore the door clear opening width in the room data sheets.
- Doorset ratings — FD30S to bedrooms, FD60S where the strategy requires.
- External wall reaction-to-fire requirements for the building height and use, per section 6.
- The separation distance between blocks that permits each to be a separate evacuation entity under 4.26.
- Confirmation of the alarm category — this document states L1 unless the strategy requires otherwise (4.15).

2.2 What a tenderer prices until it arrives

Price on this basis. State in the tender that you have done so. Do not price a worst case against it.

- Category L1 alarm to BS 5839-1 throughout.
- Simultaneous evacuation, each block a separate evacuation entity, no voice alarm — as stated at 4.26.
- REI 60 compartmentation between rooms and between room and corridor.
- FD30S bedroom doorsets, certified as doorsets.
- 1200 mm corridors, 1500 mm on accessible routes.
- Bedroom doors FAIL-SECURE, escape route doors FAIL-SAFE, per 4.18.

2.3 If the issued document differs

Where the issued strategy requires more than the above, the difference is a change under section 17 valued at the rates in item M of Appendix 1. Where it requires less, the saving is the Employer's. A tenderer that prices a worse case silently has not made an allowance, it has made itself uncompetitive.

Appendix 3 Planning consent and conditions

Authored by	The local planning authority; the site rules below are the Employer's
Status	The decision notice is to be inserted in full. Not a summary — a summary of a condition is somebody's reading of it, and the reading is what is being complied with.

3.1 What the Employer's Requirements depend on it for

- Working hours, at 7.4 and in the site rules.
- External appearance and the approved colour range at Appendix 5.
- External lighting, ecology and landscape conditions affecting the site boundary.
- The surface water discharge rate at Appendix 8.
- The duration of the consent, against the 38-month deployment and the relocation intention at 13.7.

3.2 What a tenderer prices until it arrives

Price on this basis. State in the tender that you have done so. Do not price a worst case against it.

- Working hours 07:00–19:00 Monday to Friday, 08:00–13:00 Saturday, none on Sundays or bank holidays.
- The colour range at Appendix 5 as consented.
- No external lighting spill beyond the site boundary; luminaires to be full cut-off.

3.3 If the issued document differs

Discharge of conditions is the Employer's and is scheduled at IF-12. Compliance once discharged is the Contractor's. A condition discharged late is an Employer delay under 18.3 only where the Contractor has given the early warning required at 18.5.

3.4 The Employer's site rules

These are the Employer's and are issued in full. They apply from mobilisation and they change once the first block is occupied — the rule about noise near occupied blocks is the one people forget, and it is a safety rule rather than a courtesy.

Rule	What it requires	Clause
Induction	Every person, before first entry, without exception. Project induction plus a village-specific module once the first block is occupied — the second one exists because a site where people are asleep two hundred metres away is not the same site.	14.5
Competence	Valid competence card for the work being done, carried and produced on request. A card that has expired is a card that has expired.	—
Hours	As the planning consent. Work outside them requires the Employer's prior written agreement AND the discharge of the relevant condition — the two are separate and both are needed.	7.4, IF-12
Deliveries	Booking slot from the Principal Contractor not less than 48 hours ahead. A vehicle without a slot may be turned away and the abortive cost is the Contractor's. No vehicle waits on the public highway.	14.2
Speed and traffic	10 mph throughout. No reversing without a banksman. Wheel wash before leaving. Pedestrian routes segregated and lit.	14.2
Drugs and alcohol	Testing on a for-cause and random basis. A positive test or a refusal is removal from site.	14.5
Smoking	Designated areas only. Not within 10 m of any accommodation block, at any stage, including before occupation — a habit formed during construction persists into occupation.	—
PPE	Safety footwear, hi-vis, hard hat and eye protection as a minimum, task-specific PPE per the risk assessment.	—
Permits	Required for: hot works, work at height, confined space, excavation, isolation of any live service, and any lift under 4.21. Issued by the Principal Contractor. Allow for the time in the programme rather than treating it as a delay.	14.5, 4.21

Noise near occupied blocks	Once any block is occupied, no work generating more than 65 dB(A) at the facade of an occupied block between 20:00 and 07:00 without the Employer's written agreement. Residents on nights are asleep during the working day and this is a safety matter, not a courtesy.	4.26
Photography	No photography of residents or of the interior of occupied rooms. Progress photography of construction areas is permitted and expected.	—
Parking	Limited to the spaces stated in the Employer's allocation. Transport is the Contractor's problem for any operative beyond that number. No parking on the village access road at any time — it is the emergency route.	14.4
Welfare	The Contractor provides its own to CDM 2015 Schedule 2 for the whole of its time on site. It does not use the village it is building.	14.4
Emergency	Assembly points as the fire strategy. The village alarm and the construction site alarm are separate systems and a person on site must know which one they have heard.	4.26

Appendix 4 Construction Environmental Management Plan

Authored by	The Employer's environmental consultant
Status	To be inserted. The obligations this document already imposes are listed below and stand whatever the CEMP adds.

4.1 What the Employer's Requirements depend on it for

- Hours, noise, dust and lighting controls at 10.6 and in the site rules.
- Ecological constraints, including any seasonal restriction on works near the site boundary.
- Waste, duty of care and the 90% diversion target at 10.5.
- Water discharge and pollution prevention during installation.

4.2 What a tenderer prices until it arrives

Price on this basis. State in the tender that you have done so. Do not price a worst case against it.

- 90% diversion from landfill, reported monthly, with transfer notes for every movement.
- Packaging taken back or recycled — not left as a disposal problem for the Employer.
- Dust suppression and wheel wash as the site rules.
- No seasonal restriction assumed. Where one is imposed, its programme effect is a change.

4.3 If the issued document differs

The CEMP may impose a seasonal ecological restriction on works near the boundary. It is stated here as a known unknown so that a tenderer asks rather than assumes, and so that the assumption is visible in the tender.

Appendix 5 Approved external colour and finish range

External colour and finish are a planning matter before they are an aesthetic one. The range below is the range consented under the condition at Appendix 3; a colour outside it requires the condition to be varied, which is the Employer's application to make and takes time nobody has allowed for.

Every coated metal finish shall meet the durability and colour fastness requirement at 4.22 — classification not less than RUV3 to BS EN 10169, with a warranty of not less than twenty years against a colour change exceeding ΔE 3 to BS EN 13523-3 and chalking not worse than rating 2 to BS EN 13523-10, in the exposure category at Appendix 8.

Ref	Colour	RAL	Finish	Where it may be used	Note
C-01	Anthracite grey	RAL 7016	Coated steel, matt 30% gloss	Primary external wall panel, all accommodation blocks	The consented base colour. Assume this unless the drawings say otherwise
C-02	Slate grey	RAL 7015	Coated steel, matt	Secondary panel, recesses and set-backs	
C-03	Pebble grey	RAL 7032	Coated steel, matt	Amenity and reception buildings only	Distinguishes the buildings a visitor must find
C-04	Traffic white	RAL 9016	Coated steel, satin	Soffits, reveals and window surrounds	Not permitted as a primary wall colour — reflectance was a planning objection
C-05	Moss green	RAL 6005	Coated steel, matt	Boundary-facing elevations of Blocks A and B only	Required by the landscape condition. See Appendix 3
C-06	Jet black	RAL 9005	Coated aluminium, matt	Rainwater goods, flashings and trims	
C-07	Anthracite	RAL 7016	Powder-coated aluminium, matt	Window and door frames throughout	Matched to C-01; a different anthracite is not a match
C-08	Signal red	RAL 3001	Powder-coated steel	Fire equipment, dry riser and hydrant markers only	Not decorative. Statutory identification
C-09	Natural	—	Fibre cement, through-coloured	Permitted as an alternative to C-01 where offered as a departure	Through-coloured only — a surface-coloured board fails the fastness requirement at year eight

5.1 Rules

- Roof finishes are not colour-controlled and are not visible from any consented viewpoint. Select for the twenty-year material warranty at 4.4, not for appearance.
- Internal colour is not controlled by this appendix. It is controlled by the light reflectance values in the room data sheets — walls not less than 0.5, ceilings not less than 0.7 — which are a lighting requirement and not a preference.
- Any colour offered outside this range is a departure under 13.4 and shall be priced separately, with the tenderer stating the time required for the Employer to vary the planning condition. That time is the Employer's to spend and the tenderer's to allow for.
- Colour matching between the panel, the frame and the trim is the Contractor's responsibility across all three suppliers. Three manufacturers' interpretations of RAL 7016 are three colours, and the difference is visible on a two-storey elevation in flat light.

Appendix 6 Bed demand curve, occupancy programme and sectional completion

Every number in this village comes off this curve. Beds, kitchen covers, drainage discharge, transport, laundry, catering, the size of the reception — all of it is sized from the peak, and a village sized off a peak that is really a spike is a village half empty for thirty months.

The delivery sequence at 7.2 is driven by this curve and not by manufacturing convenience. A tenderer proposing a different sequence prices it as a departure and states the effect on the first-occupation date.

6.1 Sectional completion dates

Section	Beds	Complete by	What it comprises
Section 1	65	End of month 3	Cumulative 65. First occupation: reception, one block, and the kitchen shell handed to P3.
Section 2	85	End of month 6	Cumulative 150. Amenity and laundry complete.
Section 3	55	End of month 9	Cumulative 205.
Section 4	20	End of month 12	Cumulative 225 — the full village, including all 12 RT-02 rooms.

Liquidated damages at 18.2 are £145 per bed per calendar day, applied per section from its completion date until it is certified, capped in aggregate at 7.5% of the contract sum. The build-up is at 18.2 and is stated there so the rate survives being looked at.

6.2 The curve, month by month

	Beds required	Beds available	On day shift	Event
Month 1	0	0	—	
Month 2	0	0	—	
Month 3	0	0	—	SECTION 1 COMPLETES — 65 beds available from month 4
Month 4	58	65	58 (100%)	
Month 5	62	65	62 (100%)	
Month 6	60	65	60 (100%)	SECTION 2 COMPLETES — 150 beds available from month 7
Month 7	96	150	82 (85%)	
Month 8	128	150	100 (78%)	
Month 9	141	150	102 (72%)	SECTION 3 COMPLETES — 205 beds available from month 10
Month 10	168	205	114 (68%)	
Month 11	190	205	124 (65%)	
Month 12	199	205	123 (62%)	SECTION 4 COMPLETES — 225 beds available from month 13
Month 13	214	225	128 (60%)	
Month 14	221	225	128 (58%)	
Month 15	225	225	124 (55%)	Peak
Month 16	225	225	124 (55%)	Peak
Month 17	225	225	124 (55%)	Peak
Month 18	224	225	123 (55%)	
Month 19	225	225	124 (55%)	Peak
Month 20	225	225	124 (55%)	Peak
Month 21	223	225	123 (55%)	

Month 22	225	225	124 (55%)	Peak
Month 23	225	225	131 (58%)	Peak
Month 24	220	225	132 (60%)	
Month 25	218	225	135 (62%)	
Month 26	214	225	139 (65%)	
Month 27	205	225	139 (68%)	
Month 28	196	225	137 (70%)	
Month 29	178	225	132 (74%)	
Month 30	162	225	126 (78%)	
Month 31	140	225	115 (82%)	
Month 32	121	225	104 (86%)	
Month 33	98	225	88 (90%)	
Month 34	74	225	70 (94%)	
Month 35	52	225	52 (100%)	
Month 36	31	225	31 (100%)	
Month 37	14	225	14 (100%)	
Month 38	0	225	—	

6.3 What the curve means

- **PEAK.** 225 beds, first reached in month 15 and held, within two beds, from month 15 to month 23. This matches the room schedule at section 2 exactly: 199 RT-01, 12 RT-02, 14 RT-03. The village is nonetheless sized and completed for 225 by the end of month 12, because a bed that arrives in the month it is needed has arrived late.
- **THE ACCESSIBLE ROOMS ARE NOT PROPORTIONAL.** All twelve RT-02 rooms are required by sectional completion 4. They are not distributed pro rata across the four sections, because an accessible room is required by whoever needs one, in whatever month they arrive, and the Employer cannot programme that.
- **DAY-SHIFT PROPORTION.** The last column drives Appendix 9 Criterion C. At the peak, 45% of residents are sleeping during the day. A thermal model run on a night-occupancy profile does not describe this building.
- **THE CURVE IS THE EMPLOYER'S FORECAST, NOT A GUARANTEE.** It is issued so that the Contractor sizes to it and so that the sectional completion dates can be tested against it. Variation within it is not a change; a change to the peak or to a sectional completion date is.
- **STANDING TIME.** Blocks complete against this curve stand partly occupied or unoccupied for months. That is the specific legionella risk at 4.12, and the written scheme of control shall address it rather than assume continuous occupation.

Appendix 7 Abnormal load route assessment

Authored by	The Employer's transport consultant, with the highway authority
Status	DRAFT AND INCOMPLETE. It is issued in that state deliberately. The route includes a structure subject to assessment and the assessment is not finished.

7.1 What the Employer's Requirements depend on it for

- The delivery method and vehicle configuration at 7.3.
- The module dimensions the Contractor's design produces — the route constrains the module, not the other way round, until the assessment says otherwise.
- Escort, permit and timing requirements at item D3 of Appendix 1.

7.2 What a tenderer prices until it arrives

Price on this basis. State in the tender that you have done so. Do not price a worst case against it.

- The route as drafted, with the structure available.
- Vehicle configuration and axle loading as the tenderer states with its tender, per 7.3.

7.3 If the issued document differs

THE RISK THAT THE ROUTE IS UNAVAILABLE IS THE EMPLOYER'S. It is a relevant event for an extension of time under 18.3, and it is stated as such at 11.1. The Contractor shall not assume the route is available and shall not price a contingency for it — the risk is not the Contractor's to carry and a tenderer carrying it prices itself out. What the Contractor shall do is state the vehicle configuration it needs, so that the assessment is completed against a real vehicle rather than an assumed one.

Appendix 8 Design loadings, site exposure and service supply characteristics

The design data the Contractor designs to. Where a figure below is marked TO BE CONFIRMED, the Contractor shall design to the stated assumption, state in its tender that it has done so, and price the assumption — it shall not treat the value as established. A figure that is confirmed later at a worse value is a change under section 17; a figure a tenderer assumed silently is not.

Group	Parameter	Value	Source or basis
SITE AND EXPOSURE	Site altitude above sea level	68 m	Topographical survey NR-SV-001
	Distance to shoreline	Greater than 100 km	Inland site — Terrain Category III
	Terrain category	III (country with hedges and isolated buildings)	BS EN 1991-1-4 and its UK National Annex
	Atmospheric corrosivity category	C3 (moderate)	BS EN ISO 12944-2. Drives the coating class at 4.22
STRUCTURAL — WIND	Fundamental basic wind velocity, $v(b,map)$	22.5 m/s	BS EN 1991-1-4 UK NA Figure NA.1
	Return period for the permanent condition	50 years	
	Return period, part-erected block left overnight or over a weekend	5 years for the exposure period	4.27. The Contractor states the wind speed at which a part-erected block must be tied down
	Directional and seasonal factors	To be taken as 1.0 unless justified	A reduction claimed must be shown
STRUCTURAL — SNOW AND LOADS	Characteristic ground snow load, $s(k)$	0.50 kN/m ²	BS EN 1991-1-3 UK NA for the site altitude
	Imposed floor load, bedrooms	1.5 kN/m ²	Section 6
	Imposed floor load, circulation and stairs	2.0 kN/m ²	Section 6
	Imposed floor load, amenity and dining	3.0 kN/m ²	Assembly use
	Imposed roof load, maintenance access only	0.6 kN/m ²	
	Basin fixing, vertical load at the front edge	130 kg	Room data sheets
THERMAL	External design temperature, winter	−4 °C	CIBSE Guide A, 24-hour building. Drives the heating duty in the room data sheets
	Internal design temperature, bedrooms	21 °C dry resultant	Room data sheets
	Heat-up requirement, cold start	12 °C to 21 °C within 90 minutes	Room data sheets
	Degree days, base 15.5 °C	2,180	For the operational energy declaration at 16.3
	Design weather file for the TM59 model	CIBSE DSY1, 2020 high-emissions, 50th percentile	Appendix 9
RAINFALL AND DRAINAGE	Design rainfall intensity, roof drainage	0.022 l/s per m ² (2 minute duration, 1 in 50 year)	BS EN 12056-3. Drives the sizing at 4.4
	Foul connection invert, per block	TO BE CONFIRMED — assume 1.20 m below finished floor level	P1 coordination drawing NR-TW-VIL-0203. Design to the assumption and price it
	Surface water discharge rate permitted	5 l/s per hectare	Planning condition — Appendix 3

ELECTRICAL SUPPLY	Supply characteristics at the P1 termination point	400/230 V, 50 Hz, three phase and neutral, TN-S	4.24
	Prospective fault current at the origin	TO BE CONFIRMED — assume 16 kA	DNO application in progress. 4.24
	External earth fault loop impedance, Z_e	TO BE CONFIRMED — assume 0.35 Ω	Measured and recorded per block at commissioning
	Maximum demand available, per accommodation block	TO BE CONFIRMED — assume 250 kVA	The Contractor's heating choice at 4.10 is checked against this. Reinforcement arising from its choice is its cost
	Standby generation available	One 500 kVA set under P1, for life-safety and essential loads only	The Contractor states which circuits it requires supported
WATER SUPPLY	Static head at the termination point	TO BE CONFIRMED — assume 3.5 bar	4.25
	Dynamic pressure and flow at the termination point	TO BE CONFIRMED — assume 2.0 bar at 1.5 l/s	Design to this and state the assumption. Boosting, if required, is within P2
	Fluid category at the incomer	Category 5	Backflow protection to BS EN 1717 and the Water Supply (Water Fittings) Regulations 1999
	Maximum static pressure permitted at any outlet	5 bar	Pressure reducing valves where the design exceeds it
	Incoming cold water temperature, summer design	20 °C	Drives the insulation requirement in the room data sheets
DATA AND COMMS	Fibre presented at the termination point	2 no. single-mode pairs per block	Terminated by P1; the Contractor terminates within the block
	Wireless coverage requirement	–67 dBm minimum, 2 devices per resident concurrently	Section 6
ACCESS AND LOGISTICS	Laydown area allocated to this package	0.42 ha, as shown on NR-TW-VIL-0201 Rev B	14.3. Smaller than a tenderer may assume — satisfy yourself at tender
	Crane hardstanding design bearing pressure	TO BE DESIGNED BY P1 to the pressure this Contractor states	IF-03, 4.21. Not an assumption for either party to make alone
	Working hours	As the planning consent at Appendix 3	7.4

Seven values are marked *TO BE CONFIRMED*. Design to the stated assumption, say so in the tender, and price it. A figure confirmed later at a worse value is a change under section 17; a figure assumed silently is not.

Appendix 9 Overheating criterion and the Employer's comfort brief

This village accommodates shift workers. A proportion of them sleep during the day, in a bedroom on the sunlit side of a building, in a month when the standard overheating assessment assumes the room is empty. The criterion below is written for that, because an assessment run on a default residential occupancy profile would pass a building these residents cannot sleep in.

9.1 The criterion

	Requirement	Note
Method	CIBSE TM59, dynamic thermal modelling, using the weather file at Appendix 8 (CIBSE DSY1, 2020 high-emissions, 50th percentile) for the nearest applicable location.	The model, its assumptions and its inputs are submitted before manufacture under 13.5.
Criterion A — living and amenity areas	Hours of exceedance of the adaptive comfort threshold $\Delta T \geq 1$ K shall not exceed 3% of occupied hours between 1 May and 30 September.	CIBSE TM52 Criterion 1, applied through TM59.
Criterion B — bedrooms, night	Operative temperature shall not exceed 26 °C for more than 1% of the annual hours between 22:00 and 07:00.	The standard TM59 bedroom criterion.
Criterion C — bedrooms, DAY-SLEEPING (this project)	In every room modelled as day-occupied, operative temperature shall not exceed 26 °C for more than 1% of the annual hours between 08:00 and 16:00.	THIS IS THE ADDITIONAL CRITERION. It is not in TM59 and it is required here.
Rooms to be modelled as day-occupied	Not less than 30% of RT-01 rooms, selected as the worst case for orientation and floor level, distributed across every block.	Appendix 6 shows the day-shift proportion. Modelling the best-oriented room proves nothing.
Occupancy assumption, day-sleeping rooms	One person, sleeping, 08:00 to 16:00, with the blind closed and the window closed for security and noise.	The window-open assumption that rescues most residential models is not available here — the village adjoins a live construction site.
Internal gains	TM59 residential gains, plus the actual connected load of the Contractor's own equipment selection at 4.10 and G-series items.	A model run on default gains and a design with higher-output equipment do not describe the same building.
Mechanical cooling	Not permitted as the primary means of meeting these criteria. It may be offered as a departure under 13.4, priced separately, with its energy and maintenance consequence stated for P5.	A village cooled mechanically for thirty-eight months is a different operating cost, and it is P5 that pays it.
Failure	Where the model fails any criterion, the Contractor shall change the design and re-model. It shall not adjust the occupancy, the gains or the weather file to obtain a pass.	Stated because it is the commonest way an overheating assessment is made to succeed.

Criterion C is the one that is not in TM59. It is required here because a proportion of these residents sleep between 08:00 and 16:00, and a model run on a night-occupancy profile does not describe the building they will sleep in.

9.2 The comfort brief

	Requirement	Where it comes from
Acoustic	The village adjoins a live construction site with consented working hours from 07:00. A day-sleeping resident is exposed to construction noise through the sleeping period, and the bedroom-to-outside sound insulation shall be designed for it — not merely for the bedroom-to-bedroom figure at section 6.	Design and demonstrate. This interacts with overheating: a resident who must open a window for temperature loses the acoustic performance the design assumed.
Light	Blackout to not more than 5 lux at the bed head with 20,000 lux on the external face of the glazing, per the room data sheets. A day-sleeping resident's blind is the only thing between them and the middle of the afternoon.	Room data sheets, internal blind.
Ventilation	The room shall be habitable with the window closed. Purge ventilation is a facility, not a design assumption.	Room data sheets and 4.10.

Appendix 10 Form of collateral warranty — content schedule

This is a CONTENT SCHEDULE, not a warranty. The form itself is drafted by the Employer's legal adviser and issued as an addendum before the clarification deadline. It is issued in this form now so that a tenderer can price the obligation — the number of warranties, the insurance it must carry and for how long — rather than pricing a document it has not seen.

A tenderer that requires an amendment to any clause below shall state it as a departure under 13.4 at tender. An amendment raised after award is a negotiation the Employer conducts having already lost its alternative.

Clause	What it requires	Why it is there	The negotiation to expect
Duty of care	The warrantor warrants it has exercised and will exercise the reasonable skill and care of a consultant experienced in work of a similar size, scope and complexity.	It gives the beneficiary a direct contractual route where none otherwise exists.	A warrantor may seek 'reasonable skill and care' without the experience qualifier. The qualifier is what makes the standard measurable and it should be kept.
No greater liability	The warrantor owes no greater duty to the beneficiary than it owes under the underlying contract, and may rely on any defence it would have under it.	Without it the warranty can be a larger liability than the contract it warrants, and the warrantor's insurer will say so.	This one is standard and is not usually resisted. A warranty offered without it should be looked at twice.
Net contribution	Liability is limited to the proportion that would be just and equitable having regard to the responsibility of other parties.	It is the clause the warrantor's insurer requires. Refusing it usually means no warranty at all.	Expect it. The Employer's protection is to hold warranties from every designer rather than to refuse this clause to one.
Copyright and licence	An irrevocable, royalty-free, non-exclusive licence with the right to sub-licence, for every purpose connected with the village — including relocation under 13.7 — surviving termination and not conditional on payment of any sum not already due.	The Employer intends to move this village. A licence that dies with the contract makes that impossible.	A warrantor may want the licence conditional on payment. Resist: the licence must survive a dispute, because a dispute is exactly when the Employer needs it.
Professional indemnity insurance	Not less than £5,000,000 for each and every claim, maintained for twelve years from the last sectional completion, with evidence of renewal produced annually, and immediate notice if cover ceases to be available on commercially reasonable terms.	13.6. Twelve years matches the limitation period for a deed.	A smaller warrantor may only carry £2m, or carry it in the aggregate. Decide the minimum before tender and state it, rather than discovering the position at warranty stage.
Step-in rights	The beneficiary may, on notice, step into the underlying contract in place of the Employer, and the warrantor shall not terminate without first giving the beneficiary notice and a period in which to step in.	It is what makes a warranty worth having to a funder. Without it, the funder's security is a right to sue after the project has stopped.	Warrantors accept step-in but negotiate the notice period. Twenty-one days is usual; less than fourteen is not workable.
Assignment	Assignable by the beneficiary on not fewer than two occasions without the warrantor's consent.	A village that may be sold or relocated may change hands. A warranty that cannot follow it is worth nothing to the next owner.	Two is the market position. A warrantor offering one is offering half of what is needed here.
Deleterious materials	The warrantor has not and will not specify materials known at the time of specification to be deleterious, or not in accordance with relevant British or European Standards or Agrément certificates.	It is a design warranty in a specific and testable form.	Expect a request to limit it to a named list. A list dated at execution ages badly; the general obligation does not.
Execution	As a deed, so that the limitation period is twelve years rather than six.	Six years is not long enough for a latent defect in sleeping accommodation to appear and be traced.	Not usually resisted, but it must be executed correctly — a deed executed as a simple contract is a six-year warranty nobody realises they have.

Beneficiaries required	To the Employer, and to any funder, purchaser or subsequent site owner the Employer names. Two beneficiaries assumed at tender; item A11 of Appendix 1 prices six warranties in total across the Contractor and its designing sub-contractors.	So the count is priced rather than argued about.	Where the Employer later names more beneficiaries than assumed, the additional warranties are valued under section 17 at the rate in item A11.
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This is a content schedule. The form itself is drafted by the Employer's legal adviser and issued as an addendum before the clarification deadline. Nothing in this appendix is legal advice.

Appendix 11 Asset information requirements and the data schema

The asset data the Employer needs to operate and later relocate this village. The schema is agreed under IF-14 within eight weeks of award and BEFORE data collection begins — agreeing it afterwards produces a second collection exercise at the Contractor's cost.

Delivery is as structured data. A folder of manufacturer PDFs is not an asset register; literature is required in addition, indexed against the same identifiers.

11.1 The schema

Field	Type	Format or domain	Mandatory	Example
assetId	Text	NR-P2-<block>-<floor>-<room>-<type>-<nnn>. Must match a durable physical label on the asset.	Yes	NR-P2-A-01-A114-TMV-007
classification	Text	Uniclass 2015 table Pr code	Yes	Pr_65_52_84_88
assetType	Text	From the Employer's controlled list, issued with the schema	Yes	Thermostatic mixing valve
manufacturer	Text	As on the product, not the supplier	Yes	—
model	Text	Full model reference including variant	Yes	—
serial	Text	Where the asset carries one	Where applicable	—
block	Text	A to H	Yes	A
floor	Integer	0 or 1	Yes	1
room	Text	The room number as it appears on the signage at 4.20	Yes	A114
installedDate	Date	ISO 8601, YYYY-MM-DD	Yes	2027-03-14
commissionedDate	Date	ISO 8601	Yes	2027-03-28
warrantyStart	Date	ISO 8601	Yes	2027-03-28
warrantyExpiry	Date	ISO 8601	Yes	2029-03-27
expectedLife	Integer	Years	Yes	12
maintenanceActivity	Text	The activity, not a reference to a manual	Yes	Thermal disinfection and strainer clean
maintenanceFrequency	Text	Months, or an SFG20 schedule reference	Yes	6
criticality	Integer	1, 2 or 3. 1 = an occupant cannot remain in the room while it is out of service	Yes	1
sparesReference	Text	Part number, and whether held or lead time	Where criticality = 1	TMV3-CART-15 / held
meterPoint	Text	Where the asset is metered, the point name used at 4.24 and 4.25	Where applicable	NR-P2-A-ELEC-HEAT
relocatable	Boolean	TRUE where the asset survives dismantling under 13.7	Yes	TRUE
documentRefs	Text	Semicolon-separated document numbers in the CDE	Yes	NR-P2-OM-0114; NR-P2-CERT-0231

11.2 Rules

Format	COBie 2.4 (Component, Type, System, Spare, Job sheets) or the Employer's spreadsheet issued with the schema. Not PDF. Not a proprietary export nobody else can open.
Naming and the CDE	Documents named to BS EN ISO 19650-2 and the Employer's project code, issued through the Employer's CDE. A document sent by email is not issued and does not start any review period at 13.5.

Progressive delivery	The register is populated as assets are installed, not compiled at handover. The Employer may inspect it at any time and it is a hold on payment if it is more than one month behind installation.
Meter point naming	The metering points at 4.24 and 4.25 use the same identifiers as the assets they measure, so that a consumption reading traces to an asset without a translation table.
Relocation	The relocatable flag is what makes the schedule of consumed components at 13.7 checkable. An asset marked TRUE that does not survive dismantling is a defect in the data as much as in the design.
Handover test	P5 shall be able to raise a work order against any asset, find its spare, and see its warranty position, using the register alone, on the day of handover. That is the acceptance test for this appendix.

Appendix 12 Social value framework and the travel-to-work area

Social value is 5% of the tender score. This appendix says what is measured, in what unit, and what evidence is required, so that what is scored is a commitment rather than an assertion.

Commitments offered are CONTRACTUAL. They are reported quarterly in the same units in which they were offered, and a shortfall is made good or explained. A commitment that is not measured is marketing and it will be scored as such.

The travel-to-work area for this project is defined as the area within a 25-mile road distance of the site entrance shown on NR-TW-VIL-0201. Where a commitment refers to local employment or local spend, it means within that area, evidenced by postcode.

12.1 The framework

Theme	Outcome	Measure	Unit	Evidence	Weight
Employment	Local people into work on this package	Full-time-equivalent roles filled by people resident in the travel-to-work area	FTE	Payroll postcode data, quarterly	25%
Employment	People facing barriers into work	Roles filled by people unemployed for more than six months, care leavers, ex-service personnel or prison leavers at the point of recruitment	Nr	Recruitment records and the referring organisation's confirmation	15%
Skills	Apprenticeships and structured training delivered on this package	Apprenticeship weeks and structured training weeks	Weeks	Training provider records and the apprenticeship agreement	20%
Skills	Manufacturing skills, at the place of manufacture	Training weeks delivered in the factory producing these modules	Weeks	Factory training records, and the disclosure required at 10.7	10%
Local economy	Spend with small and medium-sized enterprises	Contract value sub-let to SMEs as defined by the Companies Act 2006	% of contract value	Sub-contract records with company numbers	15%
Local economy	Spend with voluntary, community and social enterprises	Contract value spent with VCSEs	% of contract value	Sub-contract records and charity or CIC numbers	5%
The people who live here	One commitment directed at the residents of this village	Stated by the tenderer, measured in the unit it states	As offered	As offered — the tenderer defines the evidence and is held to it	10%

12.2 Rules

What is not counted	Work the Contractor would have done anyway, counted as social value. An apprentice already employed before award is not an apprenticeship created by this package, and claiming one is a scoring matter at tender and a contractual one afterwards.
Baseline	Every commitment is stated as an increment over the tenderer's existing position, and the existing position is disclosed. A percentage with no baseline is not a commitment.
Reporting	Quarterly, in the same units offered, with the evidence named in the framework. Reported to the Employer and to P5, who will be operating alongside the same community for thirty-eight months.
Shortfall	Made good within the following quarter, or explained in writing with a proposal for an equivalent. Repeated unexplained shortfall is a matter for the change and performance provisions, not a matter of goodwill.
The seventh theme	It carries a tenth of the weight and it is the one the Employer reads first. Two hundred and twenty-five people will live on this site for up to thirty-eight months, some of them a long way from home, working shifts. A commitment aimed at them is worth more than a percentage aimed at a spreadsheet.